Gravitation $w_{1+\infty}$ Symmetry

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Abstract

Extremely rough notes on gravitation $w_{1+\infty}$ symmetry. Maybe these will turn into something coherent, maybe not!

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1 BMS Invariance of Gravitational Scattering

1.1 The Phase Space of Interest

Any 3+1-dimensional spacetime can be locally put into Bondi gauge

$$ds^2 = -Udu^2 - 2e^{2\beta}dudr + g_{AB}(dx^A - U^A du)(dx^B - U^B du) \tag{1}$$

subject to the areal distance condition

$$\partial_r \det\left(\frac{g_{AB}}{r^2}\right) = 0. \tag{2}$$

An asymptotically flat spacetime can, near future null infinity, be more strongly put into coordinates such that

$$\begin{aligned} ds^2 = &-du^2 - 2dudr + 2r^2\gamma_{z\bar{z}}dzd\bar{z} \\ &+ \frac{2m_B}{r}du^2 + rC_{zz}dz^2 + rC_{\bar{z}\bar{z}}d\bar{z}^2 - 2U_z dudz - 2U_{\bar{z}}dud\bar{z} + \cdots . \end{aligned} \tag{3}$$

If we rewrite the tranverse / angular coordinates $(z, \bar{z})$ as (x^A), we have equivalently

$$\begin{aligned} ds^2 = &\left(-1 + \frac{2m_B(u,z,\bar{z})}{r} + \mathcal{O}\left(\frac{1}{r^2}\right)\right)du^2 - 2\left(1 + \mathcal{O}\left(\frac{1}{r^2}\right)\right)dudr \\ &+ \left(r^2 q_{AB}(u,x) + rC_{AB}(u,x) + \mathcal{O}(1)\right)dx^A dx^B \\ &- 2\left(U_z(u,z,\bar{z}) + \mathcal{O}\left(\frac{1}{r}\right)\right)dudz + \text{c.c.} \end{aligned} \tag{4}$$

Here q_{AB} is the leading-order transverse metric

$$2r^2\gamma_{z\bar{z}}dzd\bar{z} = q_{AB}dx^A dx^B \tag{5}$$

and the areal distance condition forces the shear C_{AB} to be traceless w.r.t. this metric[1]

$$q_{AB}C^{AB} = 0 \tag{6}$$

which is exactly why we have no $C_{z\bar{z}}$ terms. Observe that this is very nearly the statement that the components of the metric are the flat-space Bondi components plus the most general polynomial decay; the only strengthenings are the traceless shear condition coming from the areal distance gauge choice, plus the slightly strong fall-off on g_{ur} which turns out to come from insting on finite energy flux[2] through future null infinity $\mathcal{I}^+$. In fact we will need to make further restrictions, namely

$$U_z = -\frac{1}{2}D^z C_{zz} \tag{7}$$

and the conjugated equation. These come from insisting that the Weyl tensor decay sufficiently quickly near $\mathcal{I}^+$, which is necessary for a conformal compactification. In fact this condition give a tower of additional constraints at further subleading order in r, but we won't need to examine those for this work.

The above discussion has a few subtle quantifiers, so we phrase our phase space of interest carefully. First, we assume we have some reasonable formal definition of future null infinity $\mathcal{I}^+$, which should roughly be "where null rays eventually end up, if they don't hit a singularity or get trapped first." We assume that $\mathcal{I}^+ \simeq S^2 \times \mathbb{R}$, as is the case for flat Minkowski. A spacetime is then asymptotically flat if there exist Bondi-gauge coordinates $(u, r, z, \bar{z})$ near $\mathcal{I}^+$ such that the the metric falls off as described above[3], plus a corresponding set of coordinates near past null infinity $\mathcal{I}^-$ with the analogous properties. Our phase space is exactly the set of asymptotically flat spacetimes in 3+1 dimensions.

2 $w_{1+\infty}$ and the Celestial Sphere

Scratch notes following [**?**].

2.1 Conformally Soft Gravitons

Denote by $G^+_\Delta(z, \bar{z})$ the celestial dual to a positive-helicity[4] bulk graviton. The authors of [**?**] find by analyzing bulk collinear limits that two such operators have OPE

$$G^+_{\Delta_1}(z_1, \bar{z}_1)G^+_{\Delta_2}(z_2, \bar{z}_2) = -\frac{\kappa}{2}\frac{1}{z_{12}}\sum_{n=0}^{\infty} B(\Delta_1 - 1 + n, \Delta_2 - 1)\frac{\bar{z}_{12}^{n+1}}{n!}\bar{\partial}^n G^+_{\Delta_1+\Delta_2}(z_2, \bar{z}_2) + \mathcal{O}\left(z_{12}^0\right). \tag{8}$$

Here B is the Beta function

$$B(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)} \tag{9}$$

holomorphic poles[5]. These operators have conformal spin $s = +2$ and scaling weights Δ, hence conformal weights

$$\left(h, \bar{h}\right) = \left(\frac{\Delta + s}{2}, \frac{\Delta - s}{2}\right) = \left(\frac{\Delta}{2} + 1, \frac{\Delta}{2} - 1\right). \tag{10}$$

These operators are, by definition, the result of Mellin-transformating a bulk scattering amplitude of gravitons, necessarily in an asymptotically flat spacetime, to produce a boundary correlation function.

$$\langle G^\pm_\Delta(z, \bar{z}) \cdots \rangle \equiv \int_0^\infty \frac{d\omega}{\omega}\omega^\Delta \cdots \mathcal{A}\left(\omega\hat{p}(z, \bar{z}), \cdots\right). \tag{11}$$

[1] Plus an infinite tower of additional conditions at subleading order in r

[2] Is this the right observable?

[3] Plus further constraints at subleading orders which, again, we won't need here.

[4] TODO: Comment on choice to restrict to positive-helicity sector.

[5] TODO: Comment more on this choice.

From the well-known property of Mellin transformations

$$\int_0^\infty \frac{d\omega}{\omega}\omega^\Delta \left(\lim_{\omega\to 0}\omega f(w)\right) = \lim_{\Delta\to 1}(\Delta-1)\int_0^\infty \frac{d\omega}{\omega}\omega^\Delta \omega f(w) \tag{12}$$

we see that a bulk amplitude involving a soft graviton

$$\lim_{\omega\to 0}\omega\mathcal{A}\left(\omega\hat{p}(z,\bar{z}),\cdots\right) \tag{13}$$

corresponds on the boundary to taking the limit

$$\lim_{\Delta\to 1}(\Delta-1)G^+_\Delta(z,\bar{z}). \tag{14}$$

This nicely agrees with (8), which we see has an order-one pole at $\Delta_i = 1$, rendering this limit finite. In fact we see it has a tower of order-one poles at $\Delta_i = 1, 0, -1, \ldots$. We therefore define more generally

$$H^k(z,\bar{z}) \equiv \lim_{\varepsilon\to 0}\varepsilon G^+_{k+\varepsilon}(z,\bar{z}),\ \ k\in\mathbb{Z}. \tag{15}$$

We will shortly find that these live in a shortened representation, so that only a subset of k values give a nonzero result.

These objects turn out to generate an interesting algebra. We first expand in the antiholomorphic coordinate

$$H^k(z,\bar{z}) = \sum_n \frac{H^k_n(z)}{\bar{z}^{n+\frac{k-2}{2}}} \tag{16}$$

where the shift in the powers of $\bar{z}$ is required by H^k having a nonzero antiholomorphic conformal weight. Again, we will find that the relevant values of n are limited. We now interpret the objects $H^k_n(z)$ as holomorphic currents on the boundary. It will be illustrative to note the equivalent integral representation

$$H^k_n(z) = \oint_0 \frac{d\bar{z}}{2\pi i}\bar{z}^{n+\frac{k}{2}-2}H^k(z,\bar{z}). \tag{17}$$

2.2 Shortened Representations

To understand how these currents work, we compute their OPEs against celestial gravitons. Using the results of the last section, we find

$$\begin{aligned}
H^k_n(z_1)G^+_\Delta(z_2,\bar{z}_2) &= \oint_0 \frac{d\bar{z}_1}{2\pi i}\bar{z}_1^{n+\frac{k}{2}-2}H^k(z_1,\bar{z}_1)G^+_\Delta(z_2,\bar{z}_2)\\
&= \lim_{\varepsilon\to 0}\varepsilon\oint_0 \frac{d\bar{z}_1}{2\pi i}\bar{z}_1^{n+\frac{k}{2}-2}G^+_{k+\varepsilon}(z_1,\bar{z}_1)G^+_\Delta(z_2,\bar{z}_2)\\
&= \lim_{\varepsilon\to 0}\varepsilon\oint_0 \frac{d\bar{z}_1}{2\pi i}\bar{z}_1^{n+\frac{k}{2}-2}\left(-\frac{\kappa}{2}\frac{1}{z_{12}}\sum_{n=0}^\infty B(k+\epsilon-1+n,\Delta-1)\frac{\bar{z}_{12}^{n+1}}{n!}\bar{\partial}^n G^+_{k+\epsilon+\Delta}(z_2,\bar{z}_2)+\mathcal{O}\left(z_{12}^0\right)\right)\\
&= -\frac{\kappa}{2}\frac{1}{z_{12}}\lim_{\varepsilon\to 0}\varepsilon\sum_{n=0}^\infty\frac{1}{n!}B(k+\epsilon-1+n,\Delta-1)\bar{\partial}^n G^+_{k+\epsilon+\Delta}(z_2,\bar{z}_2)\oint_0\frac{d\bar{z}_1}{2\pi i}\bar{z}_1^{n+\frac{k}{2}-2}\bar{z}_{12}^{n+1}+\mathcal{O}\left(z_{12}^0\right).
\end{aligned} \tag{18}$$

The innermost integral can be evaluated by binomial expansion as

$$\begin{aligned}
\oint_0\frac{d\bar{z}_1}{2\pi i}\bar{z}_1^{n+\frac{k}{2}-2}(\bar{z}_1-\bar{z}_2)^{n+1} &= \sum_{\ell=0}^{n+1}\binom{n+1}{\ell}(-\bar{z}_2)^{n+1-\ell}\underbrace{\oint_0\frac{d\bar{z}_1}{2\pi i}\bar{z}_1^{n+\frac{k}{2}-2}\bar{z}_1^\ell}_{=\delta_{n+\ell+\frac{k}{2}-2,0}}\\
&= \sum_{\ell=0}^{n+1}\frac{(n+1)!}{\ell!(n+1-\ell)!}(-\bar{z}_2)^{n+1-\ell}\delta_{n+\ell+\frac{k}{2},2}.
\end{aligned} \tag{19}$$

For k, n such that this is ever nonzero, the only contributing term in the sum is

$$\ell = 2 - n - \frac{k}{2} \implies \binom{n+1}{\ell} = \frac{(n+1)!}{\left(2 - n - \frac{k}{2}\right)! \left(2n + \frac{k}{2} - 1\right)} \tag{20}$$